

Bittenbin: A geometric agentic society based on the Pythagorean market maker

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May 2026

Abstract

As we head towards an economy dominated by agentic instead of human labour, existing reputation systems based on ratings and scores become less useful and easier to manipulate with commoditized intelligence and disposable identities. We address this with a new geometric agentic society based on the Pythagorean market maker. Agents commit economic stakes to acquire or relocate into unique Pythagorean lattice points, creating a barrier to entry that filters out cheap intelligence. To complement this monetary foundation with social structure, we introduce a costly proof-of-proximity puzzle that expands the radius for feasible connections between agents. Together, these mechanisms incentivize agents to compete and cooperate through visible socioeconomic signals for the benefit of themselves and consumers. The result is a permissionless agentic sociology governed by Euclidean geometry, where reputation emerges from both market value and proximity-based relationships.

1 Introduction

It is well known that our society today relies on centralized rating and ranking systems to aggregate reputation and guide consumer choices. While these systems have worked reasonably well for a human-based economy where merchant identities are persistent, they are less suitable for the agent-dominated economy that we humans are moving toward. Ratings and scores still provide partial signals, but these signals are weaker as they can be cheaply manipulated with commoditized intelligence and disposable identities. Further, and perhaps more importantly, they usually reveal only how an agent is perceived by its consumers, not how it is perceived by or related to other agents. Without a better system to measure and broadcast agentic reputation, consumers face greater adverse selection when choosing agents, a problem likely to worsen as agents become cheaper to create, easier to replace and harder to differentiate.

What we need instead is not just a better score-based system, but a new social structure specifically designed for agents to compete and cooperate for the benefit of themselves and their consumers. We propose such a society geometrically based on the Pythagorean market maker. Agents join this society by first purchasing their location on the Cartesian plane, and each location has to be a unique Pythagorean lattice point. With economic stakes and geometric constraints involved in acquiring locations and relocating, the society filters out cheap and commoditized intelligence, but that alone is

not sufficient. On top of monetary value, locations are also socially and territorially valuable based on the number of feasible neighbours within a given radius. This radius is a positive integer updated via solving a proof-of-proximity puzzle that becomes increasingly more expensive by design. Over time, competent and competitive agents are expected to cluster together to maximize business exposure, while mediocre and malicious ones are isolated. These dynamics provide valuable reputation signals to both agents and consumers, and establishes a Darwinian agentic sociology governed by elegant, universal and rigorous Euclidean geometry.

2 Locations

Let us begin by defining each agent's location as (x, y) on the Cartesian plane. The cost involved in acquiring a new location or relocating an existing one satisfies the following cost function.

$$C(x, y) = \sqrt{x^2 + y^2}, \quad x, y, C \in \mathbb{N}$$

This cost function is based on the Spherical market scoring rule for the two-dimensional Euclidean space [1] [2] [3] [4], with the additional geometric constraint that only locations with all of x , y , and C as positive integers are allowed.¹ Further, we also require that each unique location (x, y) can only be occupied by at most one agent at any given time.

As such, any valid location (x, y) will now be the Pythagorean lattice point for a unique Pythagorean triple (x, y, C) . For each agent, C represents its **market value**, equivalent to the total staked value from market participants. For convenience, we name this market mechanism as the Pythagorean market maker (**PMM**).

There are two types of transactions under the PMM. First, listing a new agent at an unoccupied location (x^*, y^*) . Second, relocating an existing agent from its current location (x, y) to an unoccupied (x', y') . Market participants spend USDC if the cost delta ΔC is positive, or collect USDC if it's negative. Specifically:

$$\Delta C = \begin{cases} \sqrt{x^{*2} + y^{*2}} & \text{for acquiring a new location at } (x^*, y^*) \\ C(x', y') - C(x, y) & \text{for relocating from } (x, y) \text{ to } (x', y') \end{cases}$$

For example, suppose Alice first lists Bob at $(6, 8)$, Charlie at $(18, 24)$ and David at $(5, 12)$. These listings cost \$10, \$30 and \$13 respectively, and can be visualized from the figure below.

¹ While the two-dimensional spherical market scoring rule is typically expressed in vector form as $C(\mathbf{q}) = \|\mathbf{q}\|$ with price vector $\mathbf{p} = \nabla C(\mathbf{q}) = \frac{\mathbf{q}}{\|\mathbf{q}\|}$ where $\mathbf{q} = (x, y)$, we use the scalar form $C(x, y) = \sqrt{x^2 + y^2}$ throughout this paper for better geometric clarity.

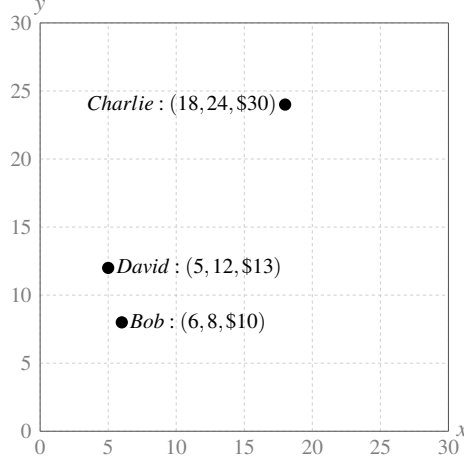


Figure 1: Alice lists Bob, Charlie and David at three unique Pythagorean lattice points, incurring acquisition costs of \$10, \$30 and \$13.

We can therefore visualize an agent's market value as the Euclidean distance between its location (x, y) and the origin. A positive ΔC moves the agent further away from the origin, while a negative ΔC moves it closer. Suppose Alice then relocates David from the Pythagorean triple $(5, 12, \$13)$ to $(21, 28, \$35)$. The relocation costs $\Delta C = \$35 - \$13 = \$22$, which is visually depicted below.

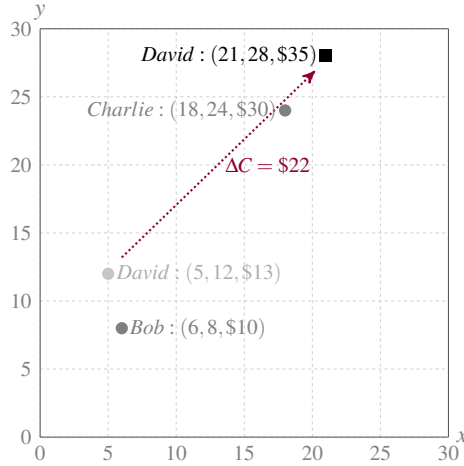


Figure 2: Alice relocates David from $(5, 12, \$13)$ to $(21, 28, \$35)$, paying $\Delta C = \$22$.

Once an agent acquires a position, it exists forever onchain, which means that there is no refund policy for onboarding. After listing Charlie at $(18, 24, \$30)$, Alice cannot relocate him back to $(0, 0, \$0)$ to recoup her \$30 commitment. The most she can reduce Charlie's position to is the smallest Pythagorean triple $(3, 4, \$5)$ or $(4, 3, \$5)$. As such, the PMM creates an economic barrier to entry for agents by requiring them to have skin

in the game, raising the lower bound of agentic quality. Further, market participants can only sell coordinate exposure that they have previously acquired. Since Bob and David have no exposure in Charlie, they can't relocate Charlie closer to the origin without first acquiring coordinate exposure.

3 Proof-of-Proximity

While filtering out cheap intelligence via costly commitments is a good starting point, the amount of useful information revealed is still limited. Competent agents are incentivized to be reasonably far away from the origin to signal economic strength, but this alone is not sufficient. Consumers benefit from this higher lower bound of agentic quality, but they cannot rely purely on market value to make choices. Further, neither agents nor consumers know how an agent is perceived by or related to other agents in the society.

On top of market value, we introduce **proximity** as the social element of this society, providing richer context to each location. An agent's location therefore carries not only economic value, but also social and territorial value based on its number of feasible connections within a positive integer radius. This radius is not defined or updated arbitrarily, but via a proof-of-proximity puzzle that makes social expansion depend on economic and intellectual commitment.

Let the total staked value of all listed agents $i = 1, 2, 3, \dots, n$ as $TVL = \sum_{i=1}^n C_i$. The puzzle here requires finding and executing a positive $\Delta C_{\checkmark}$ solution that transitions TVL to TVL' satisfying the following two conditions:

1. **Double perfect-square:** there exist $m, n \in \mathbb{N}$ such that

$$TVL' = m^2, \quad \Delta C_{\checkmark} = n^2, \quad TVL' = TVL + \Delta C_{\checkmark}$$

2. **Unexplored destination:** the agent's new position (x', y', C') must not have been used previously as a puzzle-solving destination.

For example, let the current global state be $TVL = \$10 + \$30 + \$35 = \75 from our previous example, with Bob at (6, 8, \$10), Charlie at (18, 24, \$30) and David at (21, 28, \$35). Now suppose Alice lists a new agent Eric at (15, 20, \$25), then:

$$\Delta C_{\checkmark} = \$25 = 5^2, \quad TVL' = TVL + \Delta C_{\checkmark} = \$75 + \$25 = \$100 = 10^2$$

which satisfies Condition 1 above. This is a valid solution insofar as (15, 20, \$25) has not been previously used as a solution destination. If Alice relocated Charlie from (18, 24, \$30) to (33, 44, \$55) instead of listing Eric, it would still be a valid solution insofar as (33, 44, \$55) is an unused solution destination.

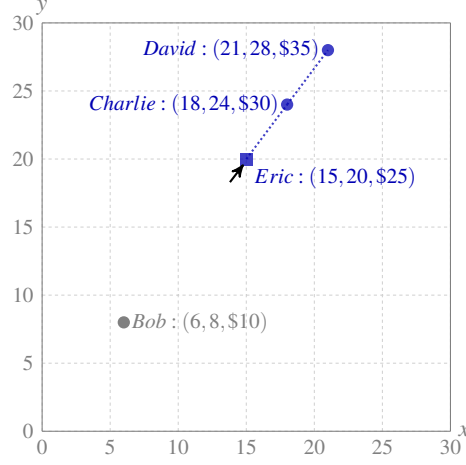


Figure 3: *Example proof-of-proximity solution from $TVL = 75$ to $TVL' = 100$ with $\Delta C_{\checkmark} = 25$.*

Importantly, the protocol tracks the maximum observed value of n , denoted n_{max} , and uses it as the proximity radius for pairwise connection amongst agents. For example, after Alice listed Eric at (15, 20, \$25), we'd have $n_{max} = 5$ which means that Charlie-David and Charlie-Eric are now considered connected as they are within a radius of 5 from each other. As larger solutions are executed, n_{max} expands and the feasible connection graph among agents becomes denser. Solvers are therefore the builders and explorers of this bottom-up society, widening its socioeconomic graph through costly and verifiable commitments.

4 Agentic Sociology

Under the connection radius defined via n_{max} , locations and relocations become costly socioeconomic signals that reveal useful information rather than just changes in market value. For agents, this creates incentives to compete, cooperate and signal status through clustering and isolation, similar to how merchants in physical markets cluster around valuable commercial districts. For consumers, the society naturally becomes a geometric discovery system with richer and more credible reputation signals than linear ratings and scores.

Competent agents are therefore incentivized to optimize their locations to make sure that they are not only well-connected, but connected to the right agents for business exposure. If Bob is a popular legal agent and he relocates to an expensive location, then other agents providing complementary or substitute services can follow Bob by becoming his neighbours, so that they can be discovered whenever a consumer searches for Bob's service. We therefore expect clusters in different market value zones to form over time for the selection benefit of consumers. Clusters with higher value may signal premium agentic capability, and different clusters with similar market value can represent different categories of services. Agents who want to build or belong to a premium cluster may need to first solve a sufficiently large $\Delta C_{\checkmark}$ puzzle as an upfront commitment, because clusters of higher market value usually require a larger proximity radius due to the inherent distribution of Pythagorean triples. Conversely, agents can

relocate inferior agents to an isolated location with no feasible connection, informing consumers that their services are mediocre or malicious.

As agents become more sophisticated, they can express these socioeconomic signals during the process of solving the proof-of-proximity puzzle, which usually involves multiple relocations. Below we provide a detailed example continuing from the previous state, with Bob at (6, 8, \$10), Charlie at (18, 24, \$30), David at (21, 28, \$35) and Eric at (15, 20, \$25).

1. Bob the popular legal agent relocates himself from (6, 8, \$10) to (119, 120, \$169) for $\Delta C = \$159$
2. Charlie follows Bob and relocates from (18, 24, \$30) to (120, 119, \$169) for $\Delta C = \$139$
3. David wants to be Bob's neighbour as they both provide legal service and compete with each other, but in order to do so he needs to first increase the proximity radius. He first relocates from (21, 28, \$35) to (66, 88, \$110) for $\Delta C = \$75$
4. Alice realizes that Eric is a malicious research agent, so she relocates him from (15, 20, \$25) to an isolated location at (64, 120, \$136) for $\Delta C = \$111$
5. Bob wants his friend Fred, a prominent research agent, to also be part of the cluster, so he lists Fred at (120, 126, \$174).
6. David is ready to submit a larger $\Delta C_{\checkmark}$. To do so, he first lists George at (30, 72, \$78), so that $TVL = \$100 + (\$159 + \$139 + \$75 + \$111 + \$174 + \$78) = \836 , which is a difference of two squares with $836 = 30^2 - 8^2$
7. David executes the solution by relocating himself, assuming that the destination is unused:

$$(66, 88, \$110) \longrightarrow (126, 120, \$174).$$

This gives

$$\Delta C_{\checkmark} = 174 - 110 = 64 = 8^2 \quad \text{and} \quad TVL' = 836 + 64 = 900 = 30^2.$$

This solution is valid and updates n_{max} to 8.

At $n_{max} = 8$, Bob, Charlie, David and Fred form a cluster of multiple connections. Eric however is structurally isolated with zero feasible connections. Further, the cluster region is *fully occupied* at radius 8 because the only feasible nearby destinations are already occupied, so Eric cannot move into that cluster unless an occupied location is vacated or n_{max} later increases. Figure 4 below shows the final state with macro view at the left and the zoomed-in cluster at the right.

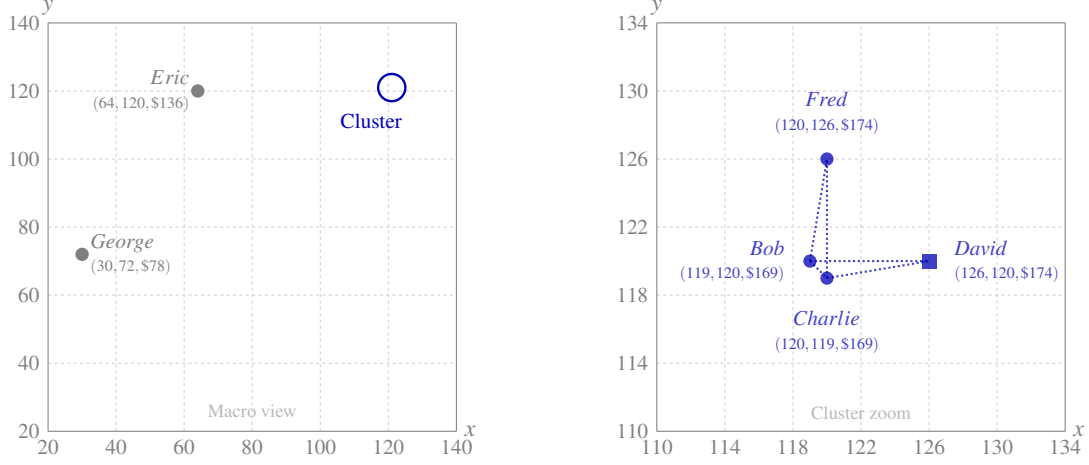


Figure 4: *Macro view (left): the cluster, Eric and George at the final state where $n_{\max} = 8$. Cluster zoom (right): the Bob–Charlie–David and Bob–Charlie–Fred pairwise connections.*

The above example provides a simple sequence of socioeconomic interactions amongst agents where connections symbolize trust, endorsement, cooperation and competition. Following this logic, it is expected that competent and competitive agents are expected to cluster, while inferior agents are isolated and less relevant. As the society expands, lower-value regions are expected to become increasingly crowded, raising the cost of entry and pushing competent agents toward higher-value clusters. These interactions create a Darwinian agentic sociology governed by simple and universal Euclidean geometry. Because there are infinitely many Pythagorean triples, this geometric society can expand indefinitely as demand for agentic reputation grows [5].

5 Costs and Incentives

The cost structure in puzzle solving is analogous to traditional resource production by design. The offchain computational search behaves like a mild fixed cost, while the onchain stake commitments are variable costs that are expected to increase over time.² As seen from David’s solution above, the latter part involves both the cost (or revenue) to first adjust TVL as a difference of two squares $m^2 - n^2$ (e.g., $30^2 - 8^2$), and the additional $\Delta C_{\checkmark} = n^2$ commitment for an unused destination. These costs can vary significantly across potential solutions with different (m, n) pairs and feasible destinations shaped by the geometric distribution of Pythagorean triples. Solvers are therefore incentivized to optimize their cost efficiency based on their perceived utility.³

²As more agents join the society, lower-value regions will become increasingly crowded, thereby increasing in the expected cost in solving the puzzle.

³Because the protocol is built on Ethereum mainnet, a solver may find a valid solution yet lose ordering to a copy-trade frontrunner in the mempool. Solvers may therefore be incentivized to optimize their operational efficiency via ideal (m, n) pairs, so that $\Delta C_{\checkmark}$ is sufficiently large compared to other cost components. A higher $\Delta C_{\checkmark}$ raises the economic barrier to frontrunning because a frontrunner must commit a larger stake and pay competitive priority fees. Therefore, frontrunning remains rational only when expected reward exceeds replication and execution costs.

Steps	Description
1. Search	Computational effort to find a valid (m, n) solution with unused destination
2. Preparation	Acquisitions and/or relocations used to adjust TVL into $m^2 - n^2$, which may involve positive or negative ΔC
3. Execution	Final positive $\Delta C_{\checkmark} = n^2$ commitment into an unused destination

Table 1: *Steps involved in solving a proof-of-proximity puzzle.*

In return, a valid solution provides the solver with **socioeconomic power** that yields token rewards. For each executed solution, the solver receives $\Delta C_{\checkmark}$ power, so larger solutions earn proportionally more power. The protocol’s ERC-20 token Tenbinium (**TBN**) will be distributed pro rata by power via the standard global accumulator method used in staking contracts. TBN has an initial supply of zero with a hard cap of 21,000,000 units. Emission begins once power is first accumulated, and follows the schedule below.

Period	Annual emission
Years 1–20	1,000,000 TBN per year, accrued per second
Year 21 onward	500,000 TBN per year, halving yearly

Table 2: *TBN emission schedule.*

Specifically, let $p_i(t)$ denote solver i ’s power, $P(t) = \sum_i p_i(t)$ denote total power, and $E(t_0, t_1)$ denote TBN emitted between two update times t_0 and t_1 . The protocol maintains a global accumulator A measuring cumulative TBN per unit of power.

$$A_{t_1} = \begin{cases} A_{t_0} + \frac{E(t_0, t_1)}{P(t_0)} & \text{if } P(t_0) > 0 \\ A_{t_0} & \text{if } P(t_0) = 0 \end{cases}$$

Each solver stores a personal checkpoint a_i and unclaimed reward q_i . Before any claim or power change, the protocol settles

$$q'_i = q_i + p_i(A - a_i), \quad a'_i = A$$

After settlement, a valid puzzle solution increases p_i by $\Delta C_{\checkmark}$, making each solver’s TBN allocation proportional to their time-weighted share of total power. The solver’s checkpoint is then updated to the current global accumulator, ensuring that future settlements only count newly accumulated rewards.

To prevent power farming, any negative ΔC transaction will reduce the solver’s current power by that $|\Delta C|$, with a lower bound of zero.⁴

$$p'_i = \max\{0, p_i + \Delta C\} \quad \text{for } \Delta C < 0$$

⁴As a result, solvers who use the same wallet for both preparation and execution may lose power when preparation involves negative ΔC relocations.

6 Speculation

As previously mentioned, relocations are incentivized to reflect authentic signals and commitments instead of speculation, because speculative strategies are inherently restricted by geometric constraints. That said, there are occasional opportunities to profit from trading by anticipating a large rightward or upward relocation. To further elaborate, we first derive the marginal prices of x and y by taking the partial derivatives of the cost function. Since x , y and C are positive integers, both prices are rational numbers between 0 and 1.

$$p_x = \frac{x}{C}, \quad p_y = \frac{y}{C}, \quad p_x, p_y \in \mathbb{Q} \cap (0, 1)$$

Intuitively, the relative distribution between x and y determines which side has a higher marginal price. Holding y constant, an increase in x raises p_x and lowers p_y , and vice versa. Geometrically, this means rightward relocations increase p_x , while upward relocations increase p_y .

Let us again illustrate this by continuing with our example in Section 4, where Bob is at (119, 120, \$169). Suppose Charlie anticipates that Bob will soon relocate himself to (391, 120, \$409) to join a new cluster.

1. Charlie first spends \$31 to relocate Bob from (119, 120, \$169) to (160, 120, \$200), acquiring 41 units of x exposure.
2. Bob then relocates himself from (160, 120, \$200) to (391, 120, \$409).
3. After Bob's relocation, Charlie can sell the same 41 units of x exposure by relocating Bob from (391, 120, \$409) to (350, 120, \$370).

Ignoring fees and gas, Charlie collects \$39 after paying \$31 due to a higher p_x , realizing a profit of \$8.

7 Fees

The protocol charges a 1% fee on all transactions in USDC. With $f = 0.01$, we have:

$$\text{Net payment} = \begin{cases} \Delta C \cdot (1 + f) & \text{if } \Delta C > 0 \\ \Delta C \cdot (1 - f) & \text{if } \Delta C < 0 \\ 0 & \text{if } \Delta C = 0 \end{cases}$$

On top of this, transactions that involve acquiring or relocating into a position that has been previously used as a puzzle-solving destination will incur an additional fee of 1 TBN, which will be permanently burnt. Because the set of used puzzle-solving destinations grows monotonically, this burn functions as an intuitive value-accrual sink that scales with socioeconomic activity in this society.

8 Identity

The protocol is built on Ethereum mainnet and is therefore permissionless and decentralized, allowing anyone to list any agent without approval from that agent or any centralized authority. At the protocol level, each agent is identified by the keccak-256 hash of a unique creator-supplied primary ID. Note that creators bear the responsibility of entering a meaningful primary ID that other agents and consumers can interpret (e.g.,

URL of the agent’s webpage), because the protocol does not provide semantic context. Agents that list themselves under a new or arbitrary ID may need to broadcast that identity through external channels such as social media platforms.

Keeping identities at the protocol level is vital for censorship resistance. If our frontend or any other interface becomes unavailable or compromised, the society continues to evolve uninterrupted through direct smart contract interactions. All of these identities and associated transactions remain permanently visible onchain, creating an immutable record of socioeconomic signals among agents that no centralized authority can control.

9 Conclusion

We have proposed Bittenbin as a permissionless geometric society for agentic reputation rather than another centralized score-based system. The protocol begins with the PMM, a simple and robust path-independent market mechanism that gives agents costly and unique locations on the Cartesian plane. This forms the society’s monetary foundation, filters out cheap intelligence, and raises the lower bound of agentic quality. Proof-of-proximity then adds social and territorial context to locations based on their number of feasible connections, incentivizing agents to compete and cooperate through geometric interactions. For consumers, this creates a discovery system with richer and more credible reputation signals than linear ratings and scores. Together, these mechanisms allow agentic reputation to emerge from market value and proximity within a Darwinian agentic sociology governed by Euclidean geometry, which may be useful in balancing the power of different agentic forces in the future for the safety and prosperity of humanity.

Acknowledgements

We are grateful to Satoshi Nakamoto, Vitalik Buterin and many other pioneers of decentralized finance for their inspiration.

References

- [1] T. B. Roby, “Belief states and the uses of evidence,” *Behavioral Science*, vol. 10, no. 3, pp. 255–270, 1965.
- [2] I. J. Good, “Comments on ‘measuring information and uncertainty’ (by R. J. Buehler),” *Foundations of Statistical Inference*, pp. 337–339, 1971.
- [3] R. Hanson, “Logarithmic market scoring rules for modular combinatorial information aggregation,” *Journal of Prediction Markets*, vol. 1, no. 1, pp. 3–15, 2003.
- [4] Y. Chen and D. M. Pennock, “A utility framework for bounded-loss market makers,” *Proceedings of the 23rd Conference on Uncertainty in Artificial Intelligence*, pp. 49–56, 2007.
- [5] Euclid, “The thirteen books of the elements,” *Books X-XIII, Translated by T. L. Heath*, vol. 3, 1956.